For full credit explain your reasoning, showing all relevant work.

Exercise 1. In the group $\mathbb{Z} \times \mathbb{Z}$ consider a subgroup D consisting of elements that have the same first and second coordinate:

$$D = \{(n, n) \mid n \in \mathbb{Z}\}$$

Since $\mathbb{Z} \times \mathbb{Z}$ is abelian, D is a normal subgroup of $\mathbb{Z} \times \mathbb{Z}$. Show that the quotient group $(\mathbb{Z} \times \mathbb{Z})/D$ is isomorphic to $\mathbb{Z}$. Hint: use the First Isomorphism Theorem.

Exercise 2. We say that an action of a group G on a set X is *transitive* if for any $x, y \in X$ there is $g \in G$ such that $g \cdot x = y$.

Show that if a finite group G acts transitively on a finite nonempty set X then

$$\sum_{x \in X} |\operatorname{Stab}(x)| = |G|$$

Exercise 3. Let G be a group such that $|G| = p^r$ for some prime p and $r \geq 1$. Assume that G acts on a set X, and that $|X| = n$ where n is not divisible by p. Show that there is an element $x_0 \in X$ such that $gx_0 = x_0$ for all $g \in G$.

PRACTICE PROBLEMS

Exercises below are for practice only - do not turn them in for grading.

Practice Exercise 1. In the group $\mathbb{Z} \times \mathbb{Z}$ consider a subgroup K defined as follows:

$$K = \{(2n, 2n) \mid n \in \mathbb{Z}\}$$

Since $\mathbb{Z} \times \mathbb{Z}$ is abelian, K is a normal subgroup of $\mathbb{Z} \times \mathbb{Z}$. Show that the quotient group $(\mathbb{Z} \times \mathbb{Z})/K$ is isomorphic to $\mathbb{Z} \times \mathbb{Z}_2$. Hint: use the First Isomorphism Theorem.

Practice Exercise 2. Let $\mathbb{R}^*$ denote the group of all non-zero real numbers with multiplication and $\mathbb{R}^+$ the group of positive real numbers with multiplication. Take the subgroup $H = \{1, -1\}$ of $\mathbb{R}^*$. Since $\mathbb{R}^*$ is an abelian group, H is a normal subgroup of $\mathbb{R}^*$. Show that $\mathbb{R}^*/H$ is isomorphic to $\mathbb{R}^+$.

Practice Exercise 3. Let G be a finite group acting on a finite set X. For $g \in G$ let $\mathrm{Fix}(g)$ be the set of elements of X which are fixed by g:

$$\mathrm{Fix}(g) = \{x \in X \mid gx = x\}$$

Show that

$$\sum_{g \in G} |\mathrm{Fix}(g)| = \sum_{x \in X} |\mathrm{Stab}(x)|$$

Practice Exercise 4. Let G be a finite group. Assume that $|G| = 45 = 5 \cdot 9$. The goal of this exercise is to show that there exists a subgroup $H \subseteq G$ such that $|H| = 9$.

Let X be the set of all subsets $A \subseteq G$ such that A consists of 9 elements. The group G acts on X as follows. If $A \in X$ and $A = \{a_1, a_2, \ldots, a_9\}$ then for $g \in G$ we define

$$g \cdot A = \{ga_1, ga_2, \ldots, ga_9\}$$

a) Show that $|X|$ (i.e. the number of elements of X) is not divisible by 3.

b) Show that there is $A_0 \in X$ such that $|\mathrm{Orb}(A_0)|$ is not divisible by 3.

c) Show that 9 divides $|\mathrm{Stab}(A_0)|$, where A_0 is as in part b).

d) Show that if $A_0 = \{a_1, a_2, \ldots, a_9\}$ and $g \in \mathrm{Stab}(A_0)$ then g must be of the form $a_i a_1^{-1}$ for some $1 \leq i \leq 9$. This shows that $|\mathrm{Stab}(A_0)| \leq 9$.

e) Conclude that $|\mathrm{Stab}(A_0)| = 9$ and so we can take $H = \mathrm{Stab}(A_0)$.